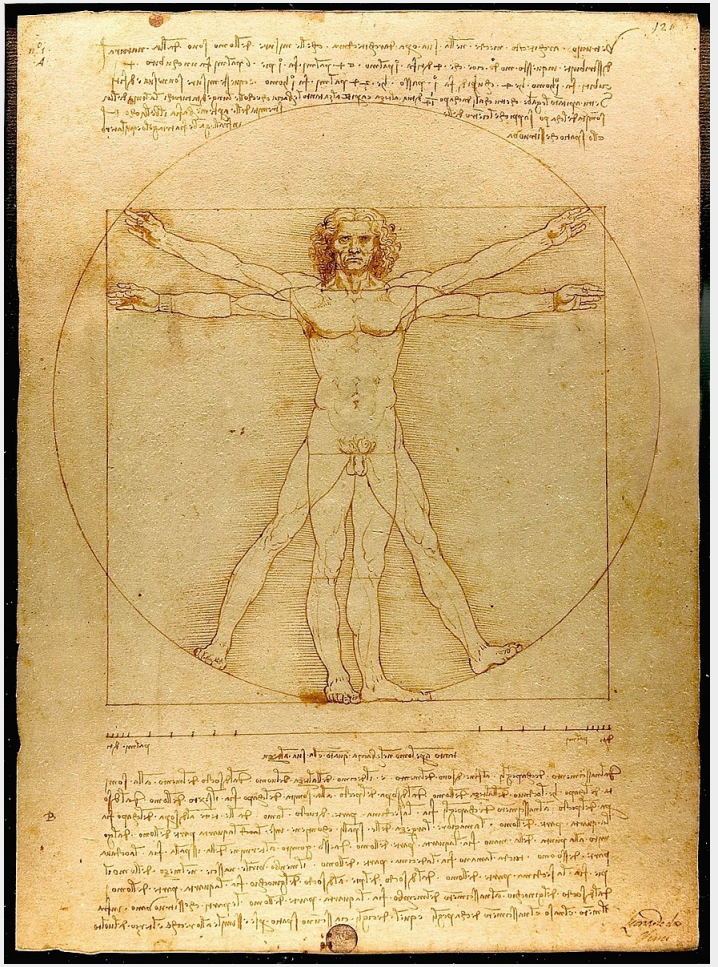
"SINTÉTICOS":

Artefatos que "CONTENHAM" INTELIGÊNCIA (?)

Gustavo Giménez-Lugo Utfpr-Dainf



CREDITS - https://en.wikipedia.org/wiki/File:Da_Vinci_Vitruve_Luc_Viatour.jpg

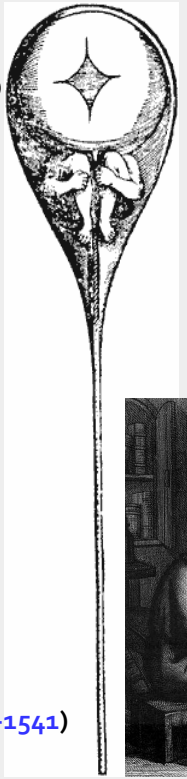
Como criar artefatos que "CONTENHAM" INTELIGÊNCIA ?

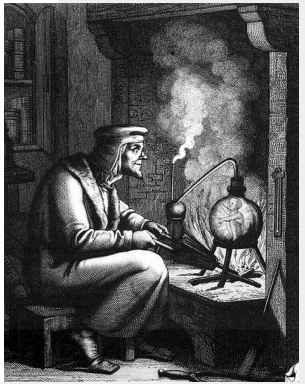
Ideias envolvendo entidades artificiais (ou passíveis de manipulação artificial)

Homúnculo



(*)Paracelso
(Phillipus Aureolus Theophrastus Bombastus von Hohenheim, 1493-1541)





O Homúnculo (Fausto II, Goethe 1832)

Golem





העוסק בספר יצירה יש לו לטהר עצמו ולגבוש
בגדים לבנים וחין לו לאדם לעסוק
יחידי כ"א בי לו ג' לכתיב ואת הנפש אשר עשו
בחרן וכתיב עובדים עבדים מן החחל וכתיב לא
עוד היות השלם לבדו לעשה לו עזר כנגדו לכך
התחיל ב' בראשית ברא . ויש לו ליקח קרקע
בתולת במקום הריס שלא הפר בת אדם סם.
ויגבל העפר במים חיים ויעשת גולם אחל ויתחיל
בגלגל בשפתי ביתות של רכ"ח שערים כל אחר
לבל כל אחר שכתיב בחו"ת בספר יצירה בגלגל
ויתגלגלו בהתהלת ח"ב וחמ"ב יגלגל בתבנית ן
ַ יַ יַ ָ וכעומם חות הבש עמהם ַ וכל השי"ב
וחמ"ב עַ וחמ"ב אַ וחמ"ב יַ וחמ"ב אָ וחמ"ב יַ וכן ח"ו
וכן ח"ה כולו וחמ"ב ימלוך ב' וכן ג' וכל אחר
בחות שבועד בו והכל יעשוק בעתרת:

Instruções para criar um Golem em hebraico, Rabino Eleazar ben Judah ben Kalonymus (1176-1238)

CREDITS (06/2009):http://pt.wikipedia.org/wiki/Homunculus
http://fotoseimagens.blogs.sapo.pt/arquivo/Rodin-1.jpg
http://commons.wikimedia.org/wiki/File:Homunculus_Faust.jpg

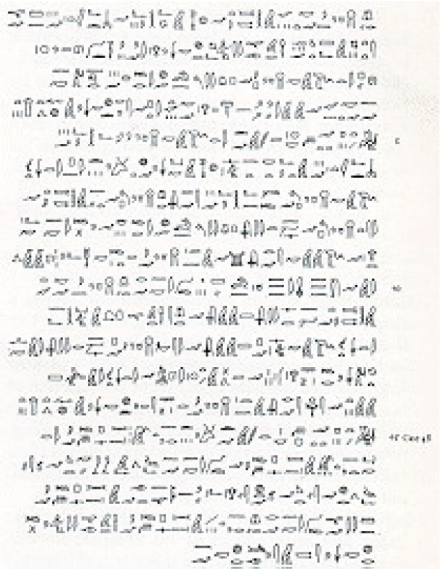
CREDITS (06/2009):http://www.newworldencyclopedia.org/entry/Golem
http://emol.org/kabbalah/golem/

Assim como artefatos para Representar CONHECIMENTO

O Papiro sobre Cirurgia "de Edwin Smith" (3000-2500 aC)

SE sintoma ENTÃO tratamento





Fonte(06/2009): http://www.neurosurgery.org/cybermuseum/pre20th/epapyrus.htm

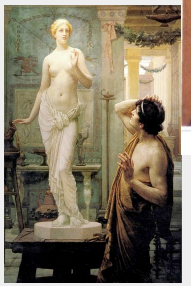
Ideias envolvendo entidades artificiais permeiam as sociedades humanas há muito tempo



Oráculos de Karnak
~1.500 a.C.
REF: https://www.jstor.org/stable/3854149?seq=1
http://www.ancient-wisdom.com/oracles.htm

Talos, o Homem de Bronze
(criação de Hefesto)
~1.000 a.C.





Pigmalião (e a escultura de mármore, animada por Afrodite)
~1.000 a.C.



Rei Mu de Zhou
registro no texto de Lieh-Tzü (Livro V, trad. de L.Giles,1912) de um autômato com forma humana criado por Yen Shih (**~1.000 a.C.**)
https://www.sacred-texts.com/tao/tt/tto8.htm
https://en.wikipedia.org/wiki/King_Mu_of_Zhou
REF: https://www.ancient-origins.net/Artifacts-ancient-technology/Advanced-technology-ancient-chinese-automata-006021



O Pombo voador
(Arquitas de Tarento, **~350 a.C**)
REF: http://kotsanas.com/gb/exh.php?exhibit=2001001

Elefante Relógio, **1200**
Isma'il Ibn al-Razzaz al-Jazari .
REF: https://upload.wikimedia.org/wikipedia/commons/7/76/Al-jazari_elephant_clock.png





Pato de Vaucanson (**1739**) (reconstrução). REF: https://vimeo.com/14904318



O Turco – **1770** (reconstrução)
REF:http://upload.wikimedia.org/wikipedia/en/2/27/Kempelen_chess1.jpg

REF:
https://www.greekmythology.com/Myths/Creatures/Talos/talos.html
https://upload.wikimedia.org/wikipedia/commons/6/6a/Pygmalion_and_Galatea_%28Normand%29.jpg

Historia de la robótica: de Arquitas de Tarento al robot Da Vinci (Parte I). Sánchez Martín FM et al. (2007). http://scielo.isciii.es/scielo.php?pid=S0210-48062007000200001&script=sci_arttext.

I propose to consider the question, "Can machines think?"

A. M. Turing (1950)
Computing Machinery and Intelligence.
***Mind 49:* 433-460.**

I propose to consider the question, "Can machines think?"
This should begin with definitions of the meaning of the terms "machine" and "think."

A. M. Turing (1950)
Computing Machinery and Intelligence.
Mind 49: 433-460.

Qual o percurso de MODELAGEM de artefatos COMPUTACIONAIS a partir de TURING ?

PARTE I

as “PARTES” da Inteligência,
uma “TEORIA GERAL”
e o CORPO como CONTENEDOR

1956 – Logic Theory Machine [Allen Newell&John Clifford Shaw&Herbert Simon]

Provador Automático de Teoremas

-provou 32 teoremas dos ***Principia Mathematica***
de Russell&Witehead

1957 – General Problem Solver [Allen Newell&Herbert Simon&John Clifford Shaw]

Raciocínio MEIOS-FINS

- sobre problemas que **PODEM SER FORMALIZADOS**
- separação de **CONHECIMENTO** e **PROCEDIMENTO/ESTRATÉGIA**
- descrição em cláusulas de Horn
- base de arquiteturas (SOAR)

Allen NEWELL & Herbert SIMON - Prêmio Turing 1975
Herbert SIMON – Prêmio Nobel de Economia 1978 (Racionalidade Limitada + "Satisfacer o suficiente" [*satisficing*: *satis*fy + su*fic*ce])
–

Nils J. Nilsson (2009). The Quest for Artificial Intelligence. Cambridge University Press.
Newell, A.; Shaw, J.C.; Simon, H.A. (1959). Report on a general problem-solving program. Proceedings of the International Conference on Information Processing.
https://psychology.wikia.org/wiki/General_Problem_Solver

1960 – O algoritmo de RESOLUÇÃO [Martin Davis&Hilary Putnam] e a UNIFICAÇÃO sintática [1965 - John Alan Robinson]

- Um "Cérebro Eletrônico" pode RACIOCINAR DEDUTIVAMENTE e para isso é suficiente DESCREVER O MUNDO

- Bastaria pensar sobre o Universo de maneira APENAS TEÓRICA

1961 – Unimate, o primeiro robô industrial[George Revol]

1965 – ELISA: gostamos de falar sobre nós mesmos [Joseph Weizenbaum]

-Ilusão de diálogo, REFLEXÃO e TEORIA DA MENTE

Weizenbaum, J.(1966). "ELIZA - A Computer Program for the Study of Natural Language Communication Between Man and Machine". Comm. of the ACM.

1966/72 – SHAKEY

SOFTWARE ARCHITECTURE

- Planner
- High Level Planning

- STRIPS (STandford Research Institute Problem Solver)
- Symbolic Planning System

- Intermediate Level Arcitecture
- for obstacle avoidance

- Low Level Actions
- for obstacle avoidance

LEVEL	FUNCTION	EXAMPLES
5 PLANEX	Executive program that calls up and monitors individual MACROPS	PLANEX might use the sensors built into the robot to determine that the block can only be fetched if SHAKEY pushes another block out of the way first – and then invoke a MACROP to fetch a block
4 STRIPS	A planning mechanism constructing MACROPS (sequences of ILAs) to carry out specific tasks	A typical MACROP might be to fetch a block from an adjacent room
3 Intermediate-level actions (ILAs)	Packages of LLAs	PUSH (OBJECT, GOAL, TOL) which instructs the robot to push a particular object to a specified goal, with a specified degree of tolerance
2 Low-level actions (LLAs)	To give the basic physical capabilities of the robot	ROLL (which tells the robot to move forward by a specified number of feet) and TILT (which tells the robot to tilt its head upward by a specified number of degrees)
1 Robot vehicle and connections to user programs	To navigate and interact physically with a realistic environment	

CREDITS:

BERMUDEZ, J.L. Cognitive Science: An Introduction to the Science of the Mind, 2014, 2nd Ed., pg 199.

http://robotglobe.org/shakey-the-worlds-first-mobile-intelligent-robot/ baseado em palestra do Peter Hart (ICRA-2015)

A VISÃO GERAL

AXIOMA BASE: uma tentativa de "TEORIA GERAL DA INTELIGÊNCIA"

Hipótese do Sistema de Símbolos Físicos (1976)

→ "Um sistema de símbolos físicos tem os meios necessários e suficientes para a ação inteligente em geral"

Influencias de:

- **Hobbes**: "raciocínio nada mais é do que cálculo"
- **Leibniz**: tentativa de criação de cálculo lógico das ideias humanas
- **Hume**: percepção poderia ser reduzida a "impressões atômicas"
- **Kant**: análise das experiências como sendo "controlada por regras formais"

CREDITS:

https://en.wikipedia.org/wiki/Physical_symbol_system

Newell, A.; Simon, H. A. (1976), "Computer Science as Empirical Inquiry: Symbols and Search", Comm. of the ACM.

A VISÃO GERAL

AXIOMA BASE: uma tentativa de "TEORIA GERAL DA INTELIGÊNCIA"

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Um computador é um sistema de símbolos físicos, como o cérebro humano e exibe propriedades iguais as do ser humano, cujo comportamento ele foi programado para simular:

- Memória de longo prazo → Conjunto de regras
- Memória de curto prazo → Memória de trabalho
- Processador cognitivo → Interpretador

REF: Newell, A.; Simon, H. A. (1976), "Computer Science as Empirical Inquiry: Symbols and Search", Comm. of the ACM.

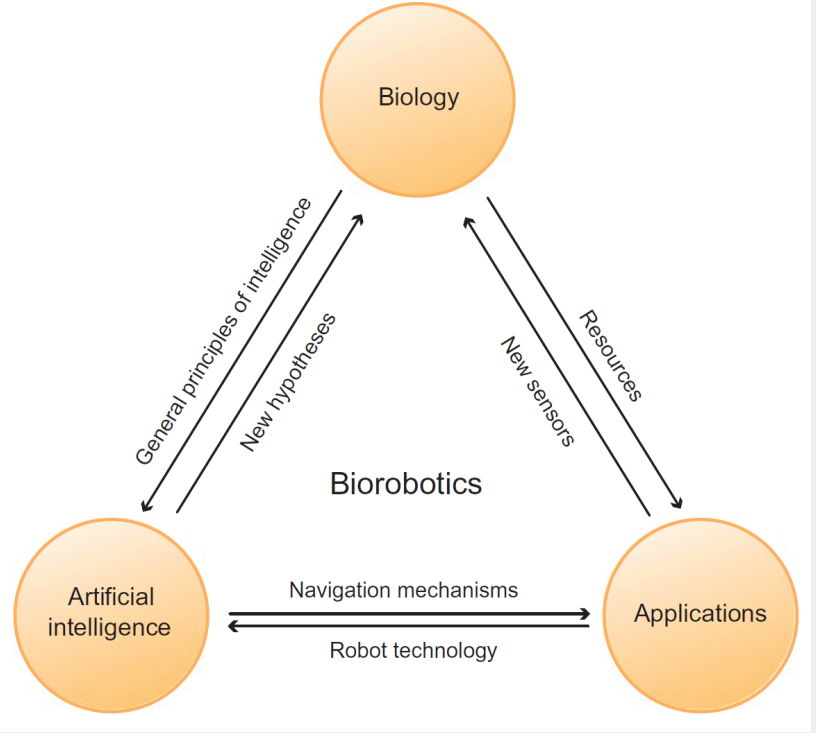
PARTE II

BEHAVIOR BASED ROBOTICS

o CORPO como "parte do" ALGORITMO
ou do "MODELO DE INTELIGÊNCIA"

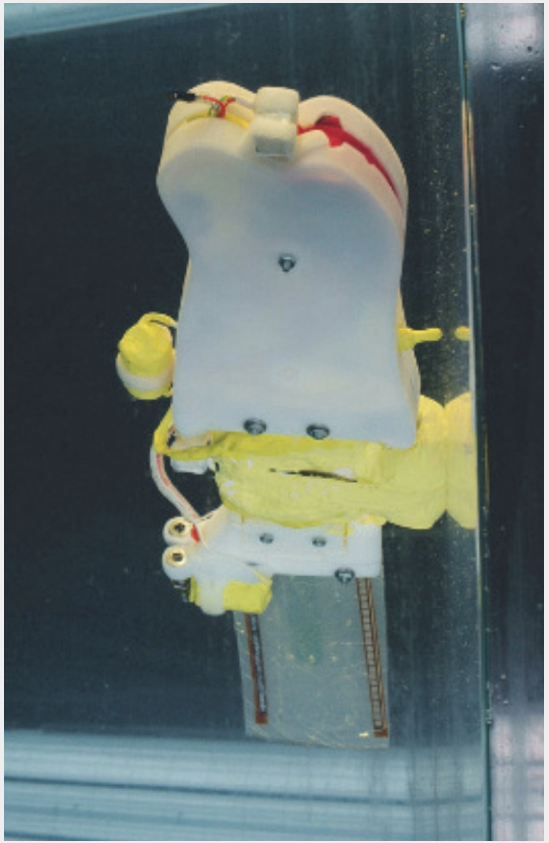
“ELEFANTES NÃO JOGAM XADREZ”

Computação Morfológica e Biorobótica



CREDITS:
BERMUDEZ, J.L. Cognitive Science: An Introduction to the Science of the Mind, 2014, 2nd Ed., pg 425

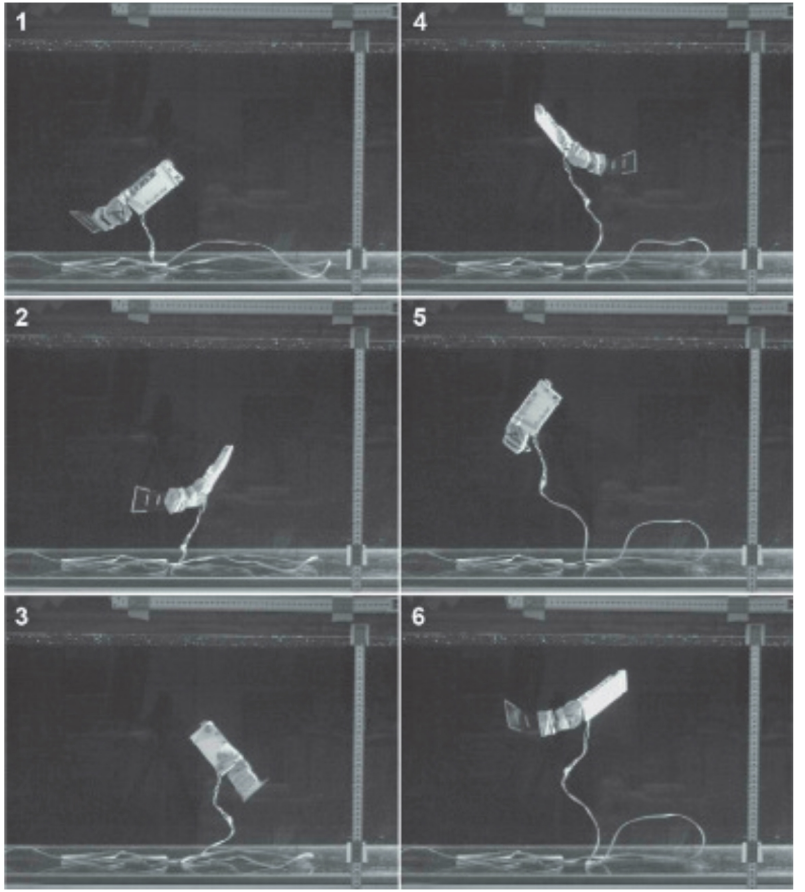
"ELEFANTES NÃO JOGAM XADREZ"



Computação Morfológica

WANDA - Peixe Robótico (Pfeifer, Lida and Gomez, 2006)

Pode apenas movimentar a cauda



CREDITS:
BERMUDEZ, J.L. Cognitive Science: An Introduction to the Science of the Mind, 2014, 2nd Ed., pg 428

“ELEFANTES NÃO JOGAM XADREZ”

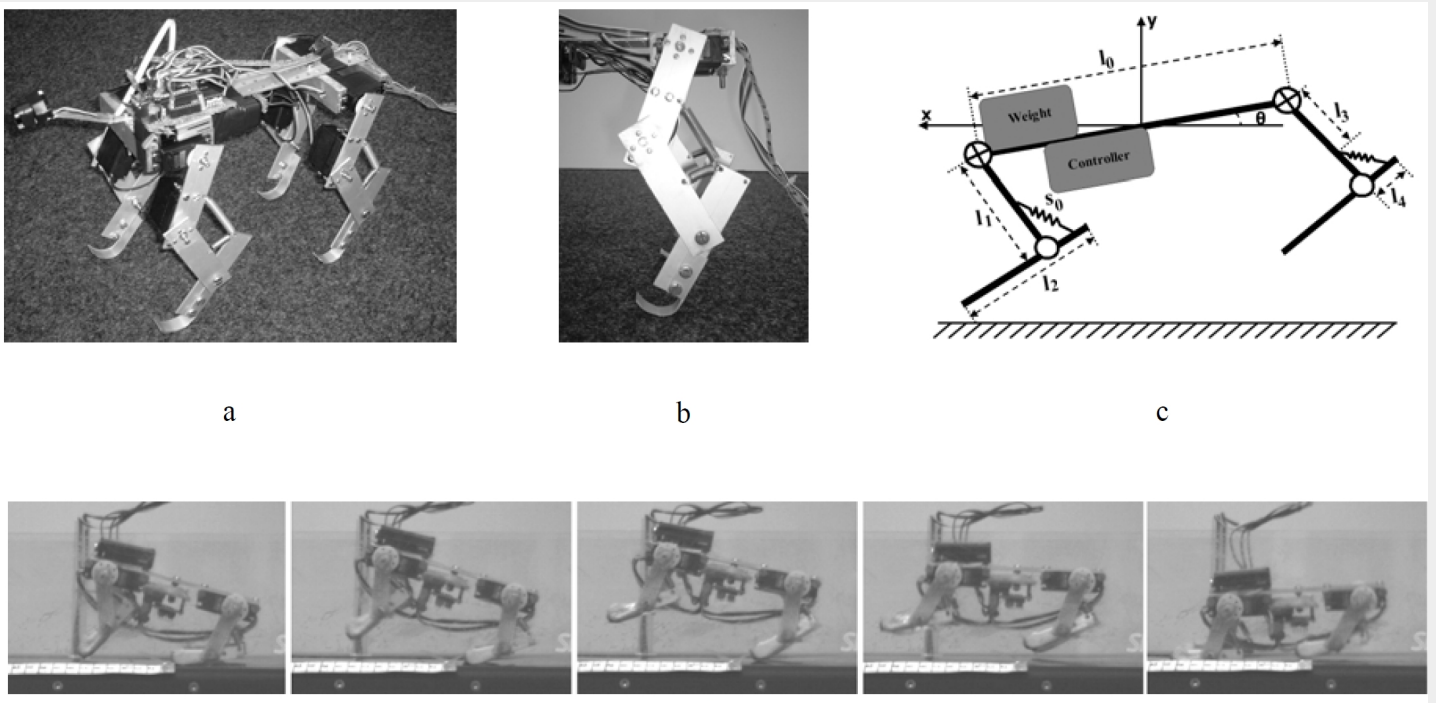


Figure 2: The quadruped “Puppy”. (a) Picture of the entire “Puppy”. (b) The spring system in the hind legs. (c) Schematic design of a slightly modified version of “Puppy . (d) Running on a treadmill.

CREDITS:

BERMUDEZ, J.L. Cognitive Science: An Introduction to the Science of the Mind, 2014, 2nd Ed., pg 428

PFEIFER,R.; IIDA, F. and GOMEZ, G. (2006). Designing intelligent robots – on the implications of embodiment Artificial Intelligence Laboratory, Department of Informatics, University of Zurich, Andreasstrasse. http://www.robotcub.org/misc/papers/07_Pfeifer_Iida_Gomez_RSJ.pdf

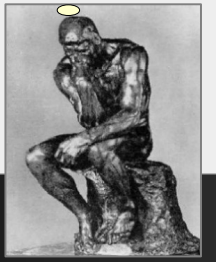
IDEM(2006). Morphological Computation: Connecting Brain, Body, and Environment. Intl. Congress Series.

continua .....



Conceptual Adaptation using image as background
Image CREDITS:
http://transitionalsupport.com.au/transitional-phase/emotional-intelligence-eq/man-thinking-hard/

Entidades (possibilidade de reflexão/ vida) X artefatos

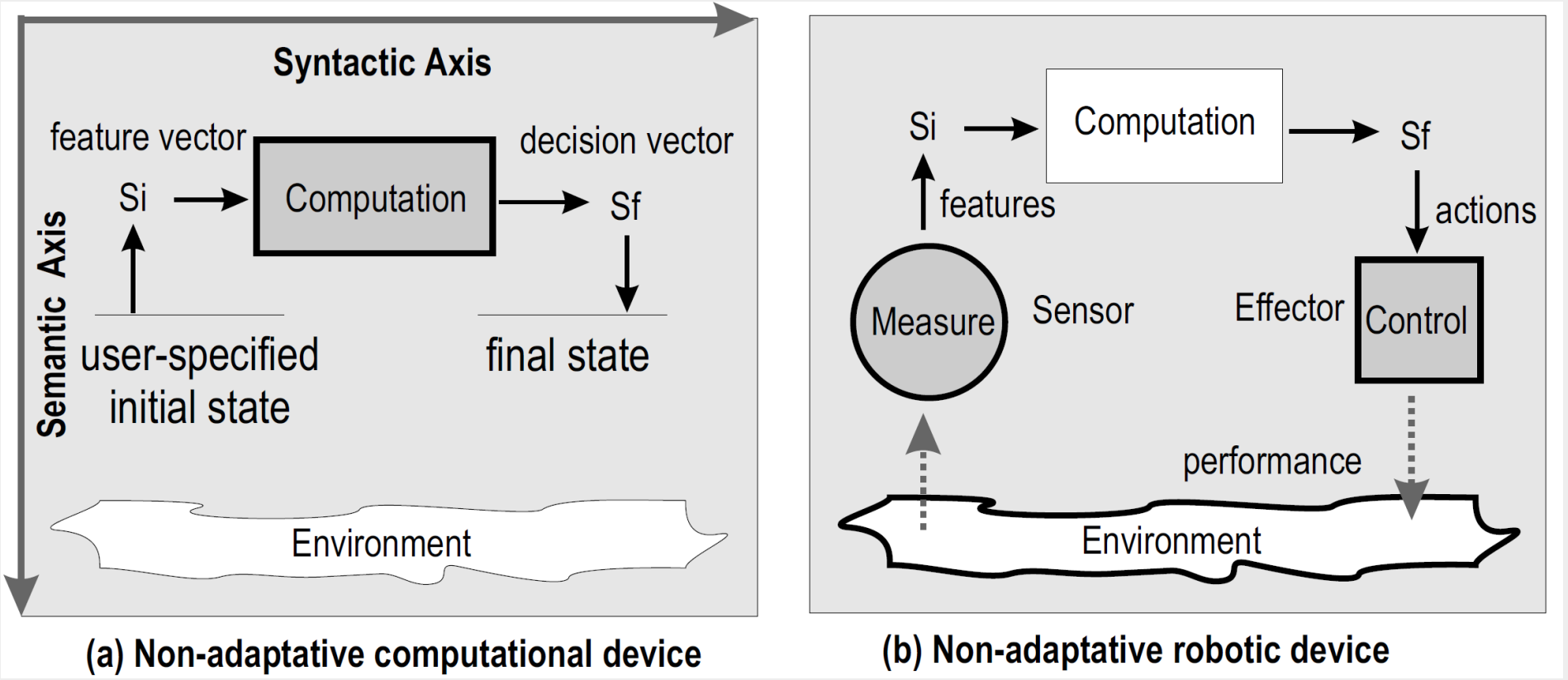


E como pensamos Modelos Computacionais

Artficiais?

Dispositivos Robóticos de Cariani

(1991a,b,c) apud Merkle, L.E. (2002)



Merkle, L.E. Disciplinary and Semiotic Relations across Human-Computer Interaction. PhDThesis, UWO, 2002.

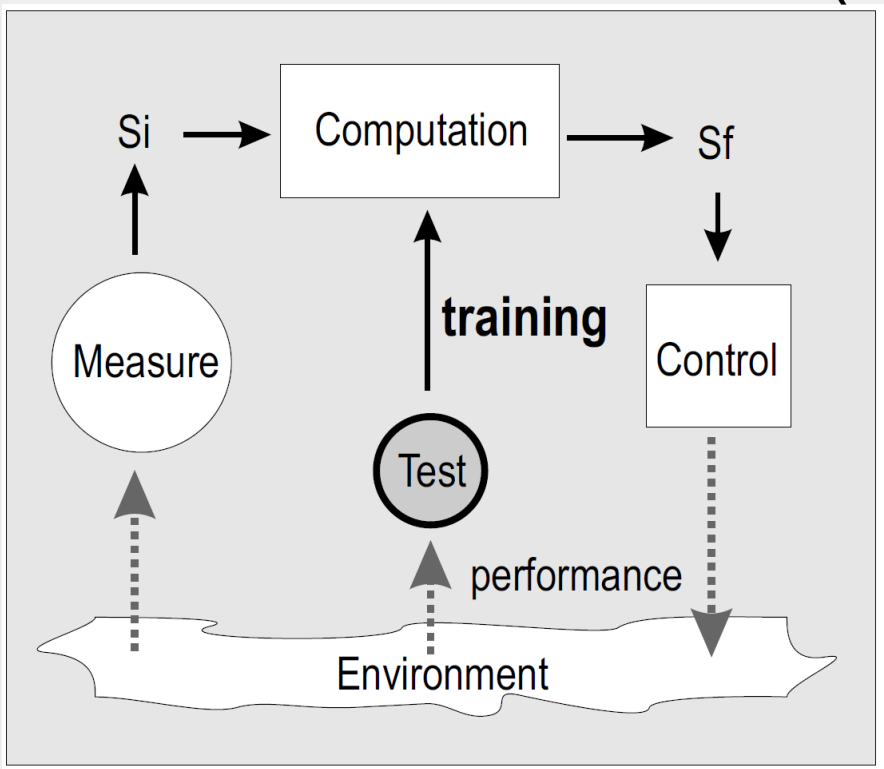
Peter Cariani. Adaptivity and emergence in organisms and devices. World Futures, 32:111-132,1991a.

Peter Cariani. Emergence and artificial life. In C.G. Langton, C. Taylor, J.D. Farmer, and S. Rasmussen, eds,. Artificial Life II , SFI Studies in the Sciences of Complexity, volume X, pages 775- 797. Addison-Wesley, 1991b.

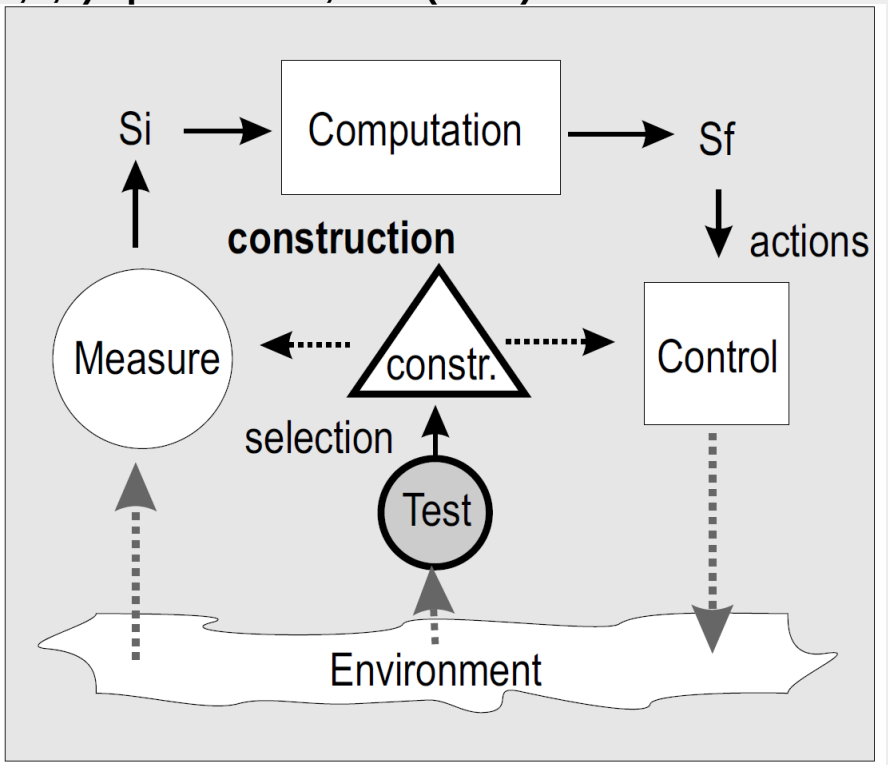
Peter Cariani. Some epistemological implications of devices which construct their own sensors and effectors. In Francisco F. Varela and P. Bourgine, editors, Proc. of the First European Workshop on Artificial Life, pages 484-493, Cambridge, MA, USA, 1991c.

Dispositivos Robóticos de Cariani

(1991a,b,c) apud Merkle, L.E. (2002)



(c) Syntactically adaptive robotic device

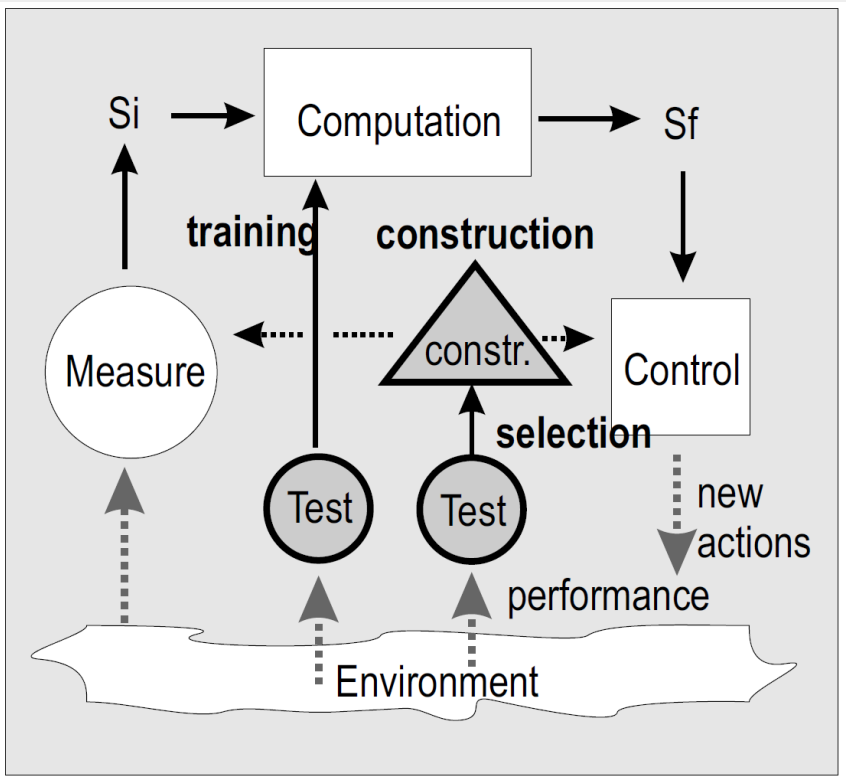


(d) Semantically adaptive robotic device

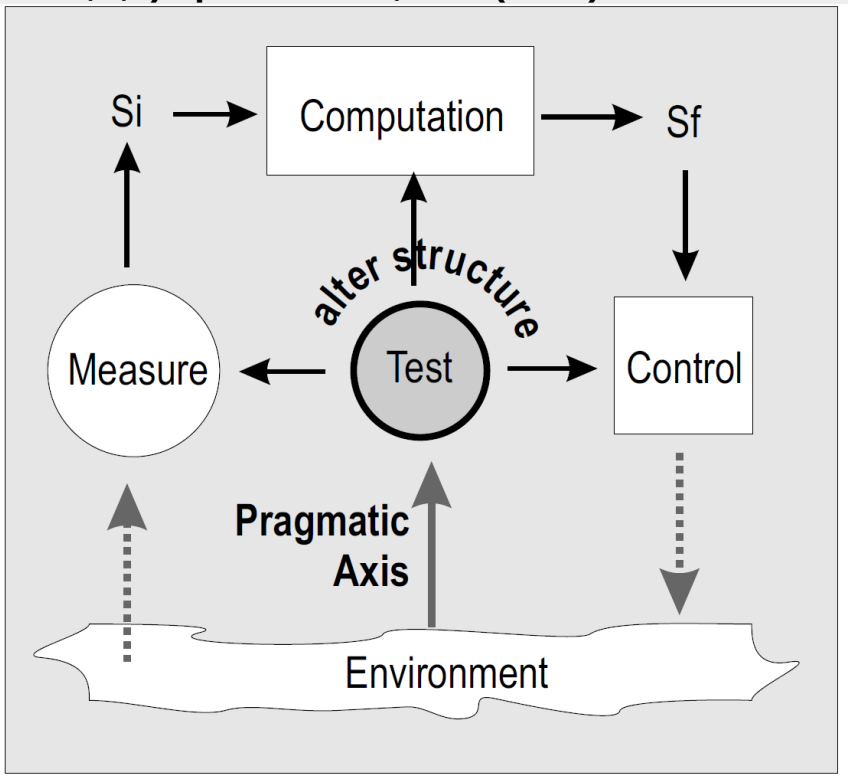
Merkle, L.E. Disciplinary and Semiotic Relations across Human-Computer Interaction. PhDThesis, UWO, 2002.
Peter Cariani. Adaptivity and emergence in organisms and devices. World Futures, 32:111-132,1991a.
Peter Cariani. Emergence and artificial life. In C.G. Langton, C. Taylor, J.D. Farmer, and S. Rasmussen, eds,. Artificial Life II , SFI Studies in the Sciences of Complexity, volume X, pages 775- 797. Addison-Wesley, 1991b.
Peter Cariani. Some epistemological implications of devices which construct their own sensors and effectors. In Francisco F. Varela and P. Bourgine, editors, Proc. of the First European Workshop on Artificial Life, pages 484-493, Cambridge, MA, USA, 1991c.

Dispositivos Robóticos de Cariani

(1991a,b,c) apud Merkle, L.E. (2002)



(e) Semantically and syntactically adaptive robotic device



(f) Evolutionary device

Merkle, L.E. Disciplinary and Semiotic Relations across Human-Computer Interaction. PhDThesis, UWO, 2002.
Peter Cariani. Adaptivity and emergence in organisms and devices. World Futures, 32:111-132,1991a.
Peter Cariani. Emergence and artificial life. In C.G. Langton, C. Taylor, J.D. Farmer, and S. Rasmussen, eds,. Artificial Life II , SFI Studies in the Sciences of Complexity, volume X, pages 775- 797. Addison-Wesley, 1991b.
Peter Cariani. Some epistemological implications of devices which construct their own sensors and effectors. In Francisco F. Varela and P. Bourgine, editors, Proc. of the First European Workshop on Artificial Life, pages 484-493, Cambridge, MA, USA, 1991c.